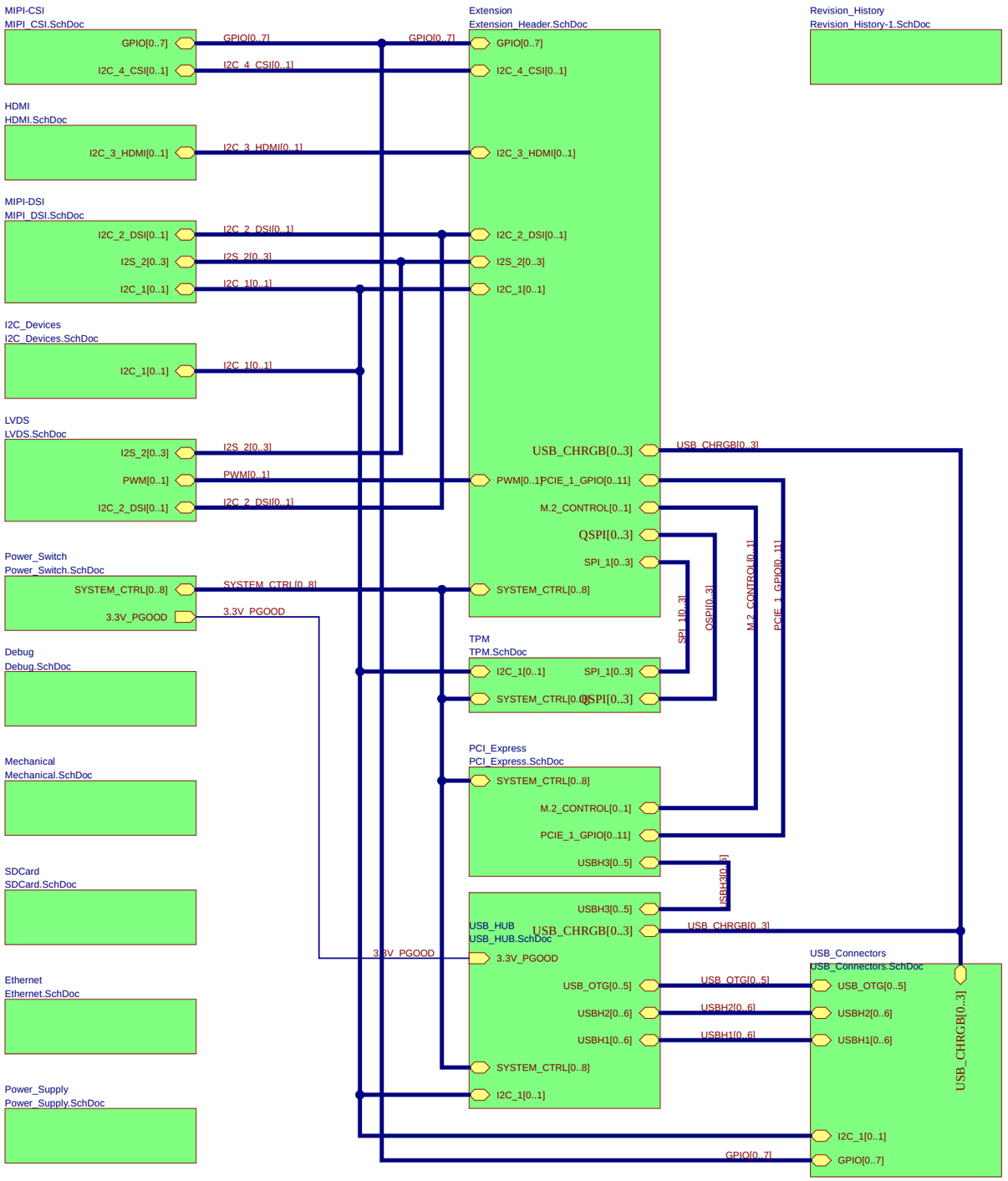


VIRGINIA INDUSTRY SOLUTIONS

Title **cb-vis-ccu**		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number:1	Revision:V2.0
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 1 of 18	
File: Hardware_Architecture.SchDoc		

REVISION HISTORY

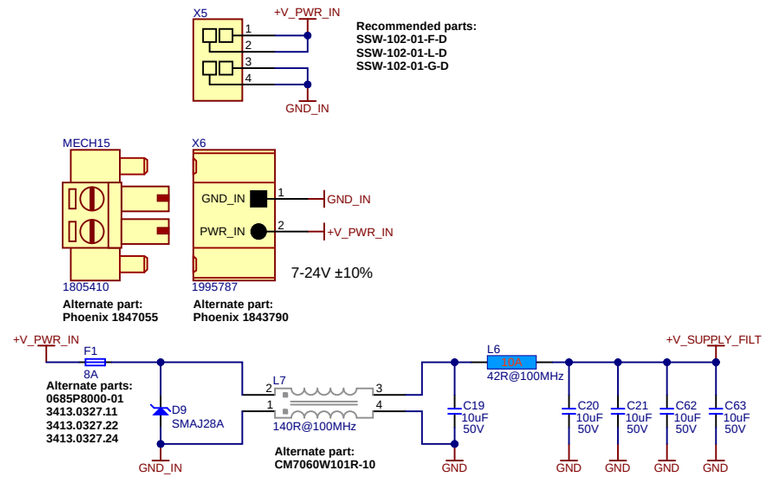
Revision V1.0

Initial release

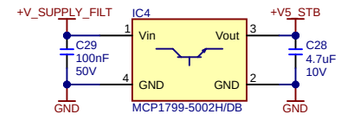
Revision V2.0
CCU (Charge controller unit) seperated from the ACDC module

Title **cb-vis-ccu**		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number:2	Revision:V2.0
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 2 of 18	
File: Revision_History-1.SchDoc		

POWER IN



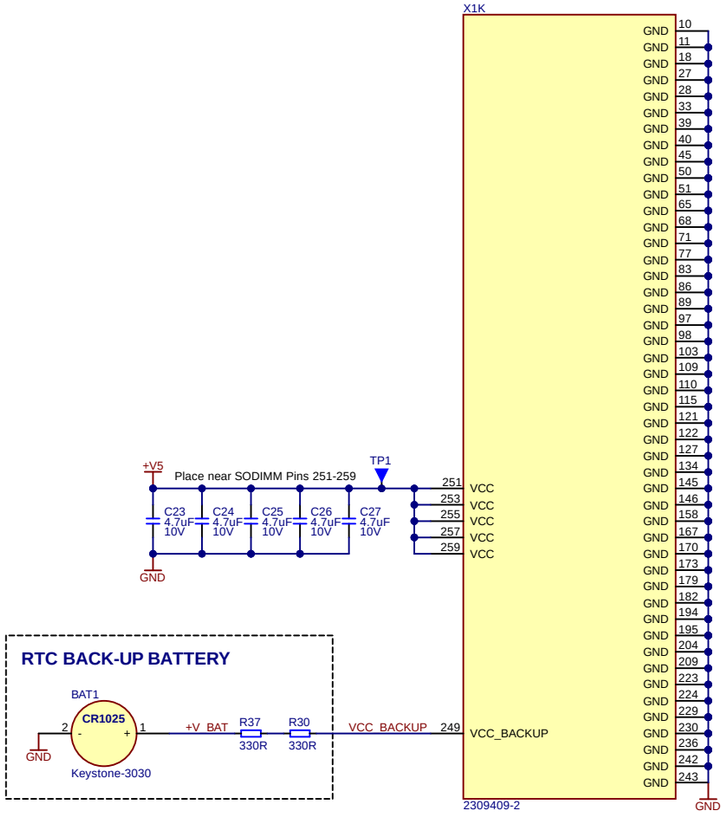
STANDBY VOLTAGE LDO



BOARD POWER RAIL SELECTION



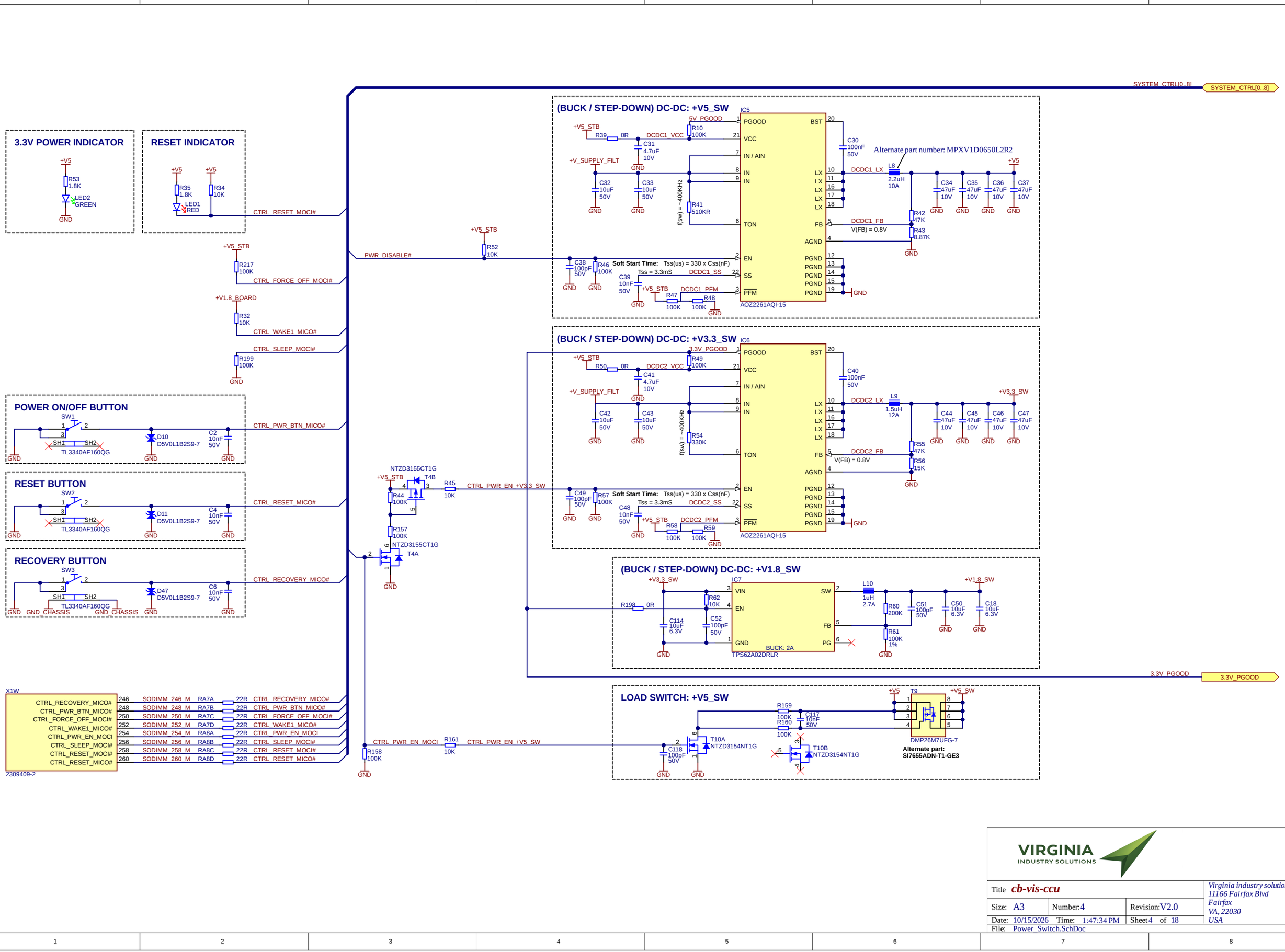
SOM POWER SUPPLY

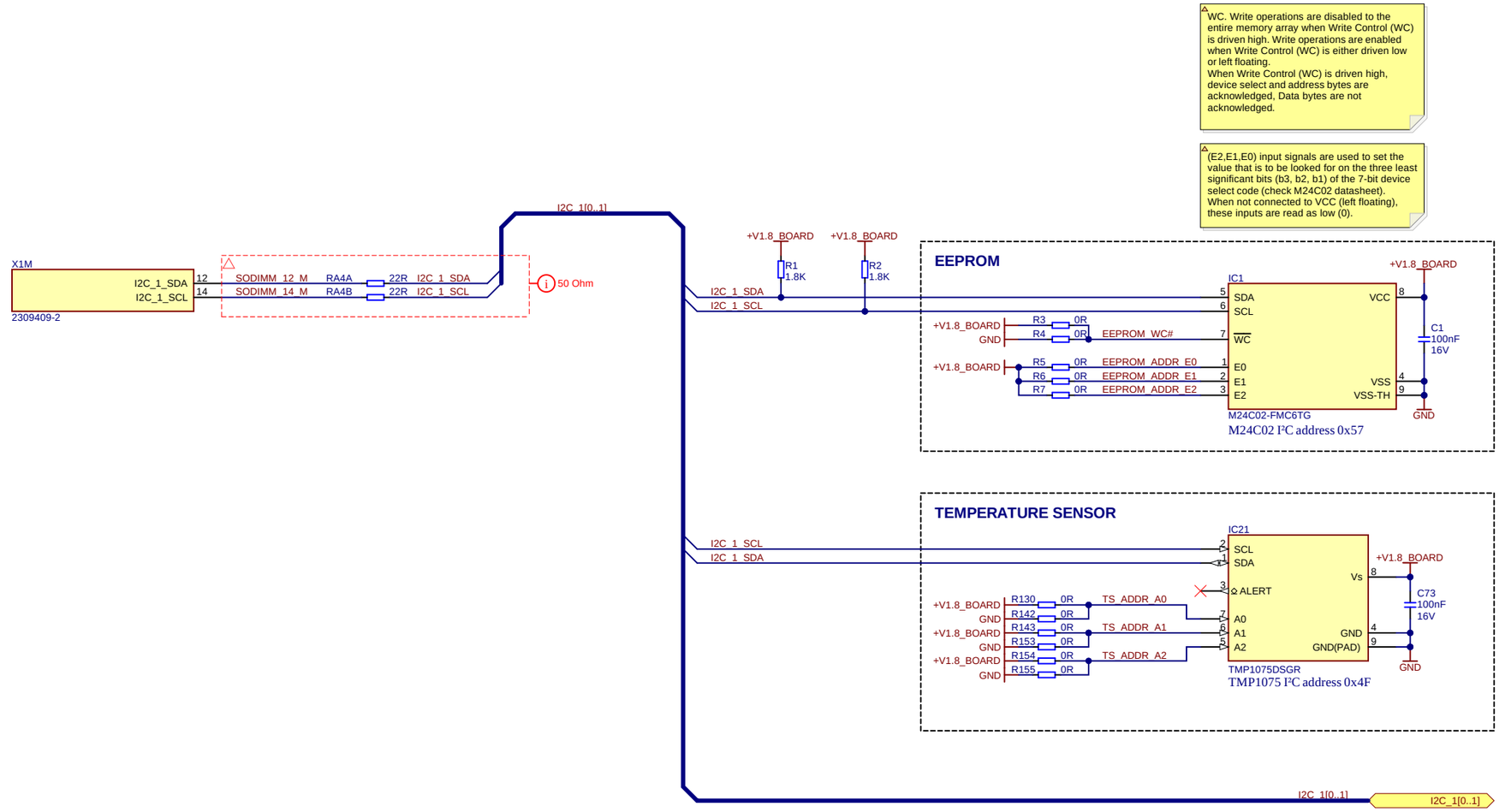


RTC BACK-UP BATTERY

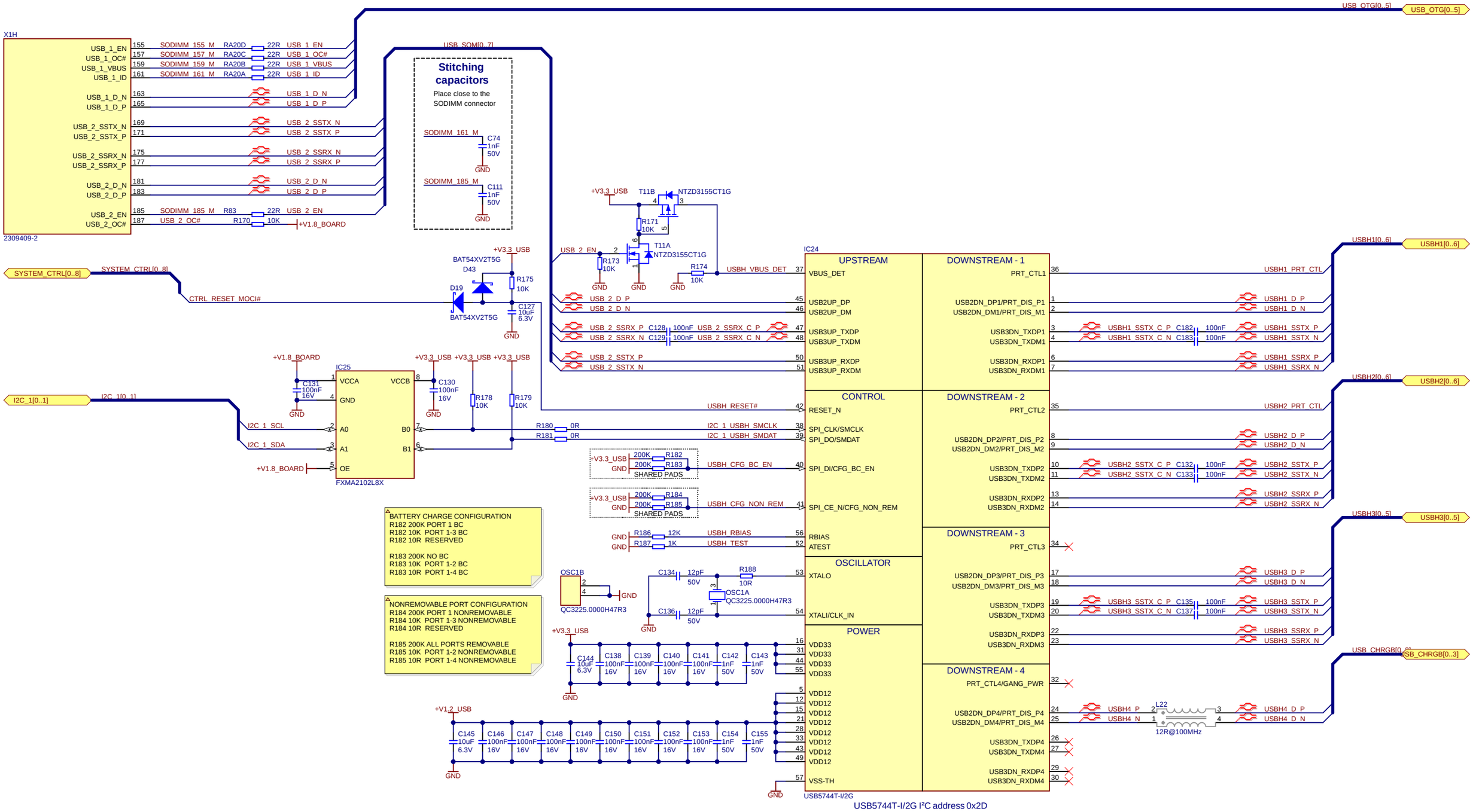
VIRGINIA INDUSTRY SOLUTIONS

Title **cb-vis-ccu**			*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number:3	Revision:V2.0	
Date: 10/15/2026 Time: 1:47:34 PM		Sheet 3 of 18	
File: Power_Supply.SchDoc			


3.3V POWER INDICATOR
+V5
R53
1.8K
LED2
GREEN
GND
RESET INDICATOR
+V5
+V5
R35
1.8K
R34
10K
LED1
RED
CTRL_RESET_MOCI#
POWER ON/OFF BUTTON
SW1
SH1
SH2
TL3340AF160QG
D10
D5V0L1B2S9-7
C2
10nF
50V
CTRL_PWR_BTN_MICO#
RESET BUTTON
SW2
SH1
SH2
TL3340AF160QG
D11
D5V0L1B2S9-7
C4
10nF
50V
CTRL_RESET_MICO#
RECOVERY BUTTON
SW3
SH1
SH2
TL3340AF160QG
GND_CHASSIS
D47
D5V0L1B2S9-7
C6
10nF
50V
CTRL_RECOVERY_MICO#
+V5_STB
R217
100K
CTRL_FORCE_OFF_MOCI#
+V1.8_BOARD
R32
10K
CTRL_WAKE1_MICO#
CTRL_SLEEP_MOCI#
R199
100K
PWR_DISABLE#
+V5_STB
R52
10K
SYSTEM_CTRL[0..8]
(BUCK / STEP-DOWN) DC-DC: +V5_SW
IC5
5V_PGOOD
+V5_STB
R39
0R
DCDC1_VCC
R10
100K
C31
4.7uF
10V
+V_SUPPLY_FILT
C32
10uF
50V
C33
10uF
50V
f(sw) = ~400KHz
R41
510KR
PGOOD
VCC
IN / AIN
IN
IN
TON
EN
SS
PFM
BST
LX
LX
LX
LX
LX
FB
AGND
PGND
PGND
PGND
PGND
PGND
C30
100nF
50V
Alternate part number: MPXV1D0650L2R2
DCDC1_LX
L8
2.2uH
10A
C34
47uF
10V
C35
47uF
10V
C36
47uF
10V
C37
47uF
10V
+V5
DCDC1_FB
V(FB) = 0.8V
R42
47K
R43
8.87K
C38
100pF
50V
R46
100K
Soft Start Time: Tss(us) = 330 x Css(nF)
Tss = 3.3mS
DCDC1_SS
C39
10nF
50V
DCDC1_PFM
R47
100K
R48
100K
AOZ2261AQI-15
(BUCK / STEP-DOWN) DC-DC: +V3.3_SW
IC6
3.3V_PGOOD
+V5_STB
R50
0R
DCDC2_VCC
R49
100K
C41
4.7uF
10V
+V_SUPPLY_FILT
C42
10uF
50V
C43
10uF
50V
f(sw) = ~400KHz
R54
330K
C40
100nF
50V
DCDC2_LX
L9
1.5uH
12A
+V3.3_SW
C44
47uF
10V
C45
47uF
10V
C46
47uF
10V
C47
47uF
10V
DCDC2_FB
V(FB) = 0.8V
R55
47K
R56
15K
C49
100pF
50V
R57
100K
Soft Start Time: Tss(us) = 330 x Css(nF)
Tss = 3.3mS
DCDC2_SS
C48
10nF
50V
DCDC2_PFM
R58
100K
R59
100K
AOZ2261AQI-15
NTZD3155CT1G
+V5_STB
T4B
R44
100K
R157
100K
NTZD3155CT1G
T4A
R45
10K
CTRL_PWR_EN_+V3.3_SW
(BUCK / STEP-DOWN) DC-DC: +V1.8_SW
+V3.3_SW
IC7
R198
0R
R62
10K
C114
10uF
6.3V
C52
100pF
50V
VIN
EN
GND
SW
FB
PG
BUCK: 2A
TPS62A02DRLR
L10
1uH
2.7A
R60
200K
R61
100K
1%
C51
100pF
50V
C50
10uF
6.3V
+V1.8_SW
C18
10uF
6.3V
3.3V_PGOOD
3.3V_PGOOD
LOAD SWITCH: +V5_SW
R159
100K
R160
100K
C117
10nF
50V
C118
100pF
50V
T10A
NTZD3154NT1G
T10B
NTZD3154NT1G
+V5
T9
+V5_SW
DMP26M7UFG-7
Alternate part:
SI7655ADN-T1-GE3
CTRL_PWR_EN_MOCI
R161
10K
CTRL_PWR_EN_+V5_SW
R158
100K
X1W
CTRL_RECOVERY_MICO#
CTRL_PWR_BTN_MICO#
CTRL_FORCE_OFF_MOCI#
CTRL_WAKE1_MICO#
CTRL_PWR_EN_MOCI
CTRL_SLEEP_MOCI#
CTRL_RESET_MOCI#
CTRL_RESET_MICO#
246 SODIMM_246_M RA7A 22R CTRL_RECOVERY_MICO#
248 SODIMM_248_M RA7B 22R CTRL_PWR_BTN_MICO#
250 SODIMM_250_M RA7C 22R CTRL_FORCE_OFF_MOCI#
252 SODIMM_252_M RA7D 22R CTRL_WAKE1_MICO#
254 SODIMM_254_M RA8A 22R CTRL_PWR_EN_MOCI
256 SODIMM_256_M RA8B 22R CTRL_SLEEP_MOCI#
258 SODIMM_258_M RA8C 22R CTRL_RESET_MOCI#
260 SODIMM_260_M RA8D 22R CTRL_RESET_MICO#
2309409-2
VIRGINIA
INDUSTRY SOLUTIONS
Title cb-vis-ccu
Size: A3
Number:4
Revision:V2.0
Date: 10/15/2026 Time: 1:47:34 PM
Sheet 4 of 18
File: Power_Switch.SchDoc
Virginia industry solutions
11166 Fairfax Blvd
Fairfax
VA, 22030
USA



Title	cb-vis-ccu		Virginia industry solutions 11166 Fairfax Blvd Fairfax VA, 22030 USA
Size: A3	Number:5	Revision:V2.0	
Date: 10/15/2026 Time: 1:47:34 PM		Sheet 5 of 18	
File: I2C_Devices.SchDoc			


X1H
USB_1_EN
USB_1_OC#
USB_1_VBUS
USB_1_ID
USB_1_D_N
USB_1_D_P
USB_2_SSTX_N
USB_2_SSTX_P
USB_2_SSRX_N
USB_2_SSRX_P
USB_2_D_N
USB_2_D_P
USB_2_EN
USB_2_OC#
2309409-2
USB_OTG[0..5]
USB_SOM[0..7]
Stitching capacitors
Place close to the SODIMM connector
SODIMM_161_M
C74 1nF 50V
SODIMM_185_M
C111 1nF 50V
GND
+V3.3_USB
T11B NTZD3155CT1G
T11A NTZD3155CT1G
R171 10K
R173 10K
R174 10K
USB_2_EN
USBH_VBUS_DET
SYSTEM_CTRL[0..8]
CTRL_RESET_MOCI#
BAT54XV2T5G
D43
D19
R175 10K
C127 10uF 6.3V
IC24
UPSTREAM
VBUS_DET
USB2UP_DP
USB2UP_DM
USB3UP_TXDP
USB3UP_TXDM
USB3UP_RXDP
USB3UP_RXDM
DOWNSTREAM - 1
PRT_CTL1
USB2DN_DP1/PRT_DIS_P1
USB2DN_DM1/PRT_DIS_M1
USB3DN_TXDP1
USB3DN_TXDM1
USB3DN_RXDP1
USB3DN_RXDM1
USBH1[0..6]
USBH1_PRT_CTL
USBH1_D_P
USBH1_D_N
USBH1_SSTX_C_P_C182 100nF
USBH1_SSTX_C_N_C183 100nF
USBH1_SSTX_P
USBH1_SSTX_N
USBH1_SSRX_P
USBH1_SSRX_N
CONTROL
RESET_N
SPI_CLK/SMCLK
SPI_DO/SMDAT
SPI_DI/CFG_BC_EN
SPI_CE_N/CFG_NON_REM
RBIAS
ATEST
USBH_RESET#
I2C_1_USBH_SMCLK
I2C_1_USBH_SMDAT
USBH_CFG_BC_EN
USBH_CFG_NON_REM
USBH_RBIAS
USBH_TEST
R186 12K
R187 1K
DOWNSTREAM - 2
PRT_CTL2
USB2DN_DP2/PRT_DIS_P2
USB2DN_DM2/PRT_DIS_M2
USB3DN_TXDP2
USB3DN_TXDM2
USB3DN_RXDP2
USB3DN_RXDM2
USBH2[0..6]
USBH2_PRT_CTL
USBH2_D_P
USBH2_D_N
USBH2_SSTX_C_P_C132 100nF
USBH2_SSTX_C_N_C133 100nF
USBH2_SSTX_P
USBH2_SSTX_N
USBH2_SSRX_P
USBH2_SSRX_N
DOWNSTREAM - 3
PRT_CTL3
USB2DN_DP3/PRT_DIS_P3
USB2DN_DM3/PRT_DIS_M3
USB3DN_TXDP3
USB3DN_TXDM3
USB3DN_RXDP3
USB3DN_RXDM3
USBH3[0..5]
USBH3_D_P
USBH3_D_N
USBH3_SSTX_C_P_C135 100nF
USBH3_SSTX_C_N_C137 100nF
USBH3_SSTX_P
USBH3_SSTX_N
USBH3_SSRX_P
USBH3_SSRX_N
DOWNSTREAM - 4
PRT_CTL4/GANG_PWR
USB2DN_DP4/PRT_DIS_P4
USB2DN_DM4/PRT_DIS_M4
USB3DN_TXDP4
USB3DN_TXDM4
USB3DN_RXDP4
USB3DN_RXDM4
USB_CHRGB[0..3]
USBH4_P
USBH4_N
L22 12R@100MHz
USBH4_D_P
USBH4_D_N
OSCILLATOR
XTALO
XTALI/CLK_IN
OSC1B
QC3225.0000H47R3
OSC1A
C134 12pF 50V
C136 12pF 50V
R188 10R
POWER
VDD33
VSS-TH
C144 10uF 6.3V
C138 100nF 16V
C139 100nF 16V
C140 100nF 16V
C141 100nF 16V
C142 1nF 50V
C143 1nF 50V
+V1.2_USB
C145 10uF 6.3V
C146 100nF 16V
C147 100nF 16V
C148 100nF 16V
C149 100nF 16V
C150 100nF 16V
C151 100nF 16V
C152 100nF 16V
C153 100nF 16V
C154 1nF 50V
C155 1nF 50V
USB5744T-I/2G
USB5744T-I/2G I²C address 0x2D
+V1.8_BOARD
IC25
VCCA
VCCB
GND
A0
A1
OE
B0
B1
FXMA2102L8X
C131 100nF 16V
C130 100nF 16V
R178 10K
R179 10K
I2C_1[0..1]
I2C_1_SCL
I2C_1_SDA
R180 0R
R181 0R
R182 200K
R183 200K
R184 200K
R185 200K
SHARED PADS
R186 12K
R187 1K
USB_2_SSTX_P
USB_2_SSTX_N
USB_2_SSRX_P C128 100nF USB_2_SSRX_C_P
USB_2_SSRX_N C129 100nF USB_2_SSRX_C_N
USB_2_D_P
USB_2_D_N
SODIMM_155_M RA20D 22R USB_1_EN
SODIMM_157_M RA20C 22R USB_1_OC#
SODIMM_159_M RA20B 22R USB_1_VBUS
SODIMM_161_M RA20A 22R USB_1_ID
USB_1_D_N
USB_1_D_P
USB_2_SSTX_N
USB_2_SSTX_P
USB_2_SSRX_N
USB_2_SSRX_P
USB_2_D_N
USB_2_D_P
SODIMM_185_M R83 22R USB_2_EN
USB_2_OC# R170 10K +V1.8_BOARD
BATTERY CHARGE CONFIGURATION
R182 200K PORT 1 BC
R182 10K PORT 1-3 BC
R182 10R RESERVED
R183 200K NO BC
R183 10K PORT 1-2 BC
R183 10R PORT 1-4 BC
NONREMOVABLE PORT CONFIGURATION
R184 200K PORT 1 NONREMOVABLE
R184 10K PORT 1-3 NONREMOVABLE
R184 10R RESERVED
R185 200K ALL PORTS REMOVABLE
R185 10K PORT 1-2 NONREMOVABLE
R185 10R PORT 1-4 NONREMOVABLE
USB POWER SWITCH
3.3V_PGOOD
IC19
VIN
EN
GND
SW
FB
PG
BUCK: 2A
TPS62A02DRLR
L35 1uH 2.7A
R200 10K
R201 0R
C158 10uF 6.3V
C90 100pF 50V
R191 100K 1%
R195 100K 1%
C157 100pF 50V
C156 10uF 6.3V
C178 10uF 6.3V
C159 10uF 6.3V
R192 10K
R193 10K
C160 100nF 16V
R197 10K
T13 MMBT5551M3T5G
T12 PMPB33XP,115
Alternate part : 2SC6026MFVGR,L3F
VIRGINIA INDUSTRY SOLUTIONS
Title cb-vis-ccu
Size: A3
Number:6
Revision:V2.0
Date: 10/15/2026 Time: 1:47:34 PM
Sheet 6 of 18
File: USB_HUB.SchDoc
Virginia industry solutions
11166 Fairfax Blvd
Fairfax
VA, 22030
USA

IC10 PORT PIN CONFIGURATION

Tri-level input pin to indicate port mode. The state of this pin is sampled when EN_N isasserted low in the TUSB320L device, EN is asserted high in the TUSB320H device, and VDD is active. This pin is also sampled following a I2C_SOFT_RESET.

H - DFP (Pull-up to Vif DFP mode is desired)
NC - DRP (Leave unconnected if DRP mode is desired)
L - UFP (Pull-down or tie to GND if UFP mode is desired)

IC10 ADDR MODE PIN CONFIGURATION

C address or GPIO mode:

H — I2C is enabled and I2C 7-bit address is 0x67.
NC — GPIO mode (IC is disabled)
L — I2C is enabled and IC 7-bit address is 0x47.

ADDR pin should be pulled up to VDD

Alternate parts:
TPS2553DRVT
TPS2553DRVR
NCP380HMUAJAATBG
NCV380HMUAJAATBG
MP62551DGT

Bleeding resistor

Alternate parts:
PI5USB30216CXUAEX
PTN5150HXMP
FUSB303BTMX

Check part's datasheet first

RC-delay circuit used to delay EN signal for at least 50 ms

Alternate parts:
TPS2553DRVT
TPS2553DRVR
NCP380HMUAJAATBG
NCV380HMUAJAATBG
MP62551DGT

Alternate parts:
TPS2553DRVT
TPS2553DRVR
NCP380HMUAJAATBG
NCV380HMUAJAATBG
MP62551DGT

Alternate parts:
692141030100 (with EMI springs)
KUSBX-AS2N9-1-BL30
KUSBX-AS2N9-2-BL30

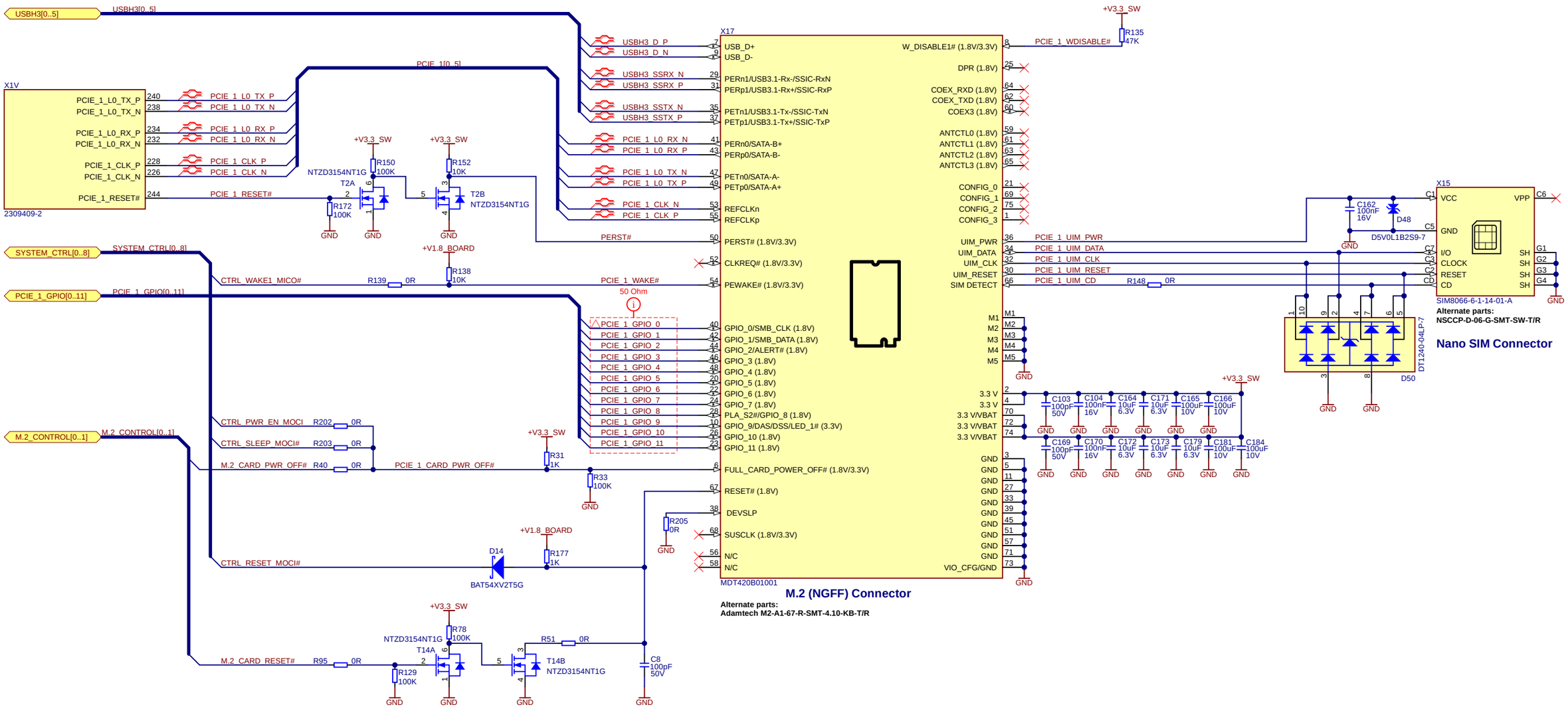
Alternate part:
12402012E212A

USB Type-C DRP Connector (Source only)

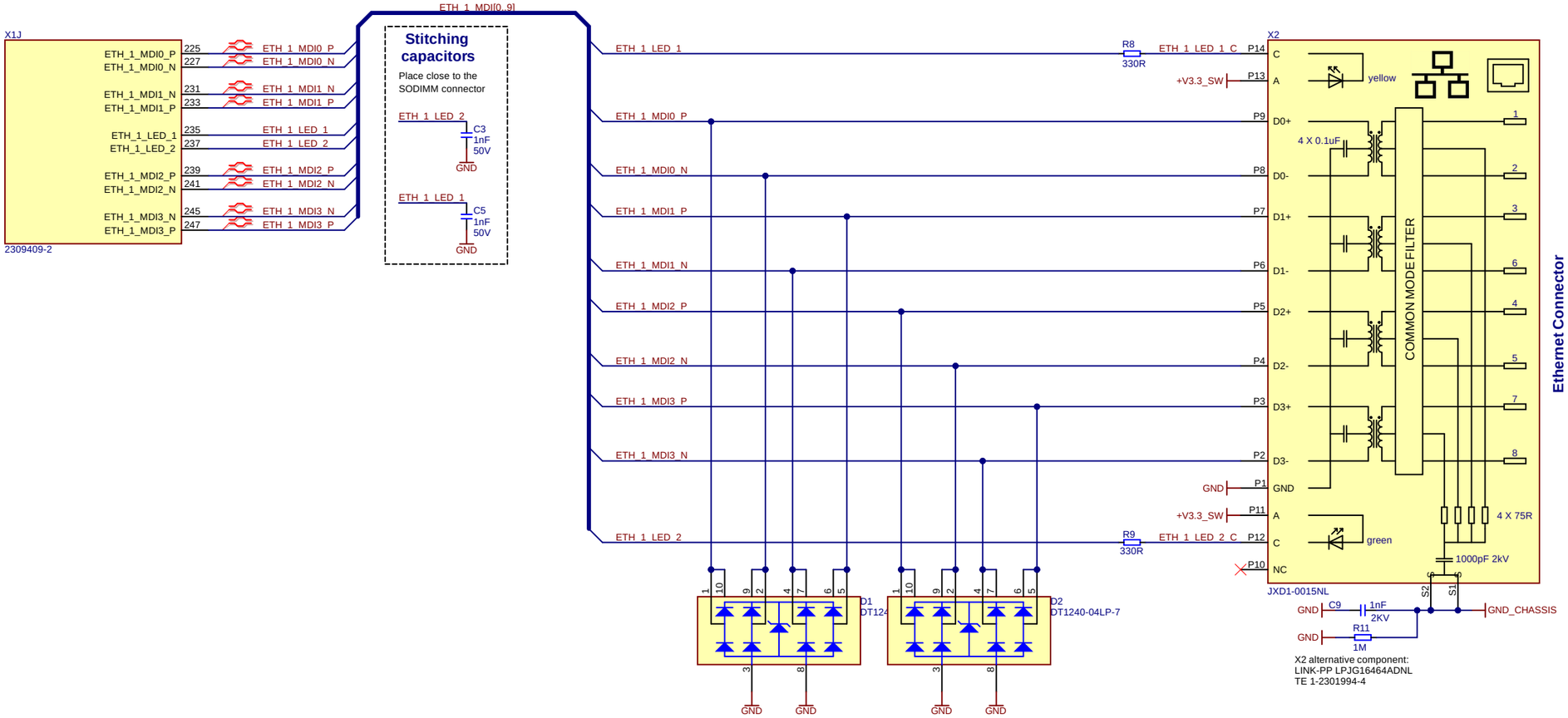
2x USB Type-A Host Connector

Title	cb-vis-ccu
Size: A3	Number:7 Revision:V2.0
Date: 10/15/2026	Time: 1:47:34 PM Sheet 7 of 18
File:	USB_Connectors.SchDoc

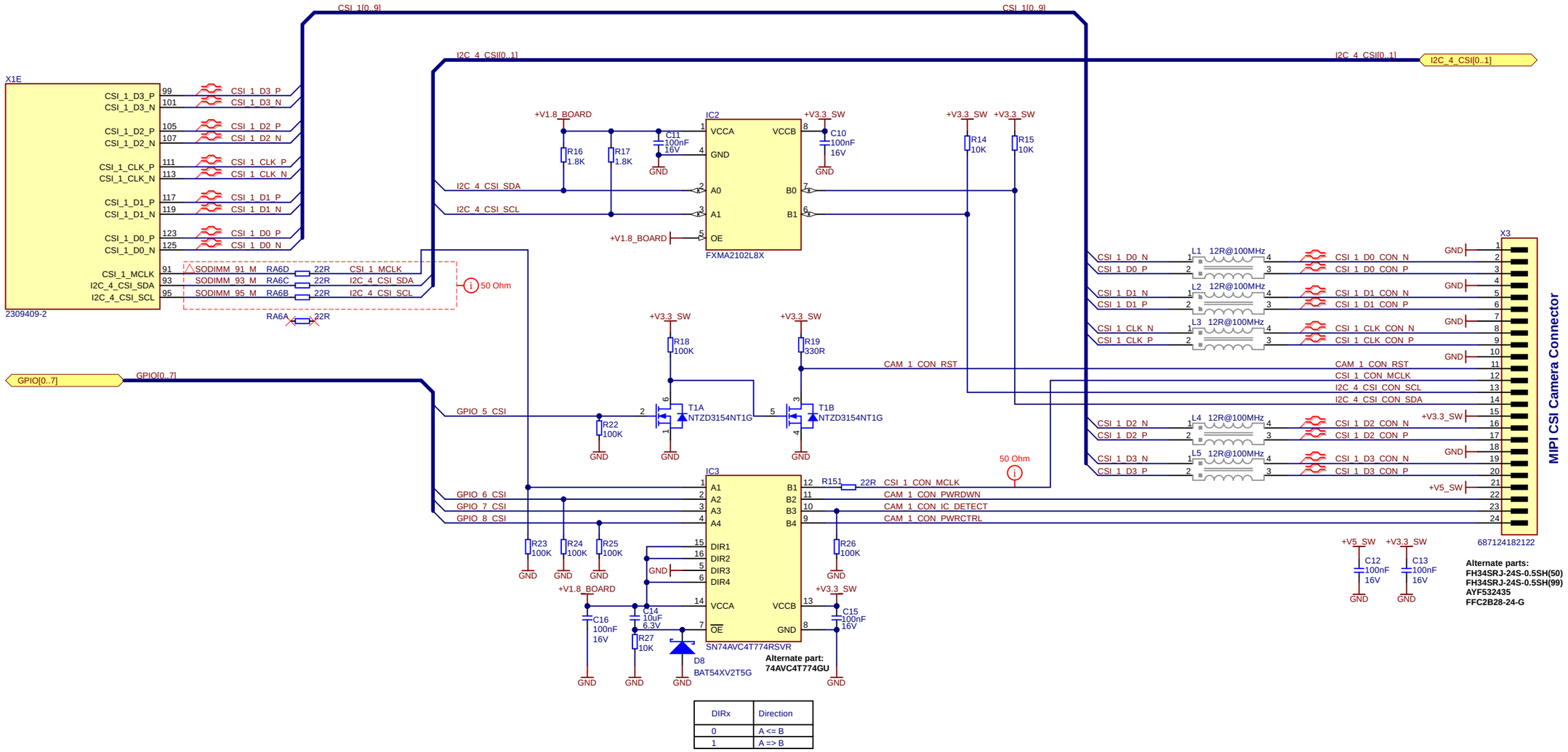
Virginia industry solutions
11166 Fairfax Blvd
Fairfax
VA, 22030
USA



Title	cb-vis-ccu		Virginia industry solutions 11166 Fairfax Blvd Fairfax VA, 22030 USA
Size: A3	Number:8	Revision:V2.0	
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 8 of 18		
File: PCI_Express.SchDoc			



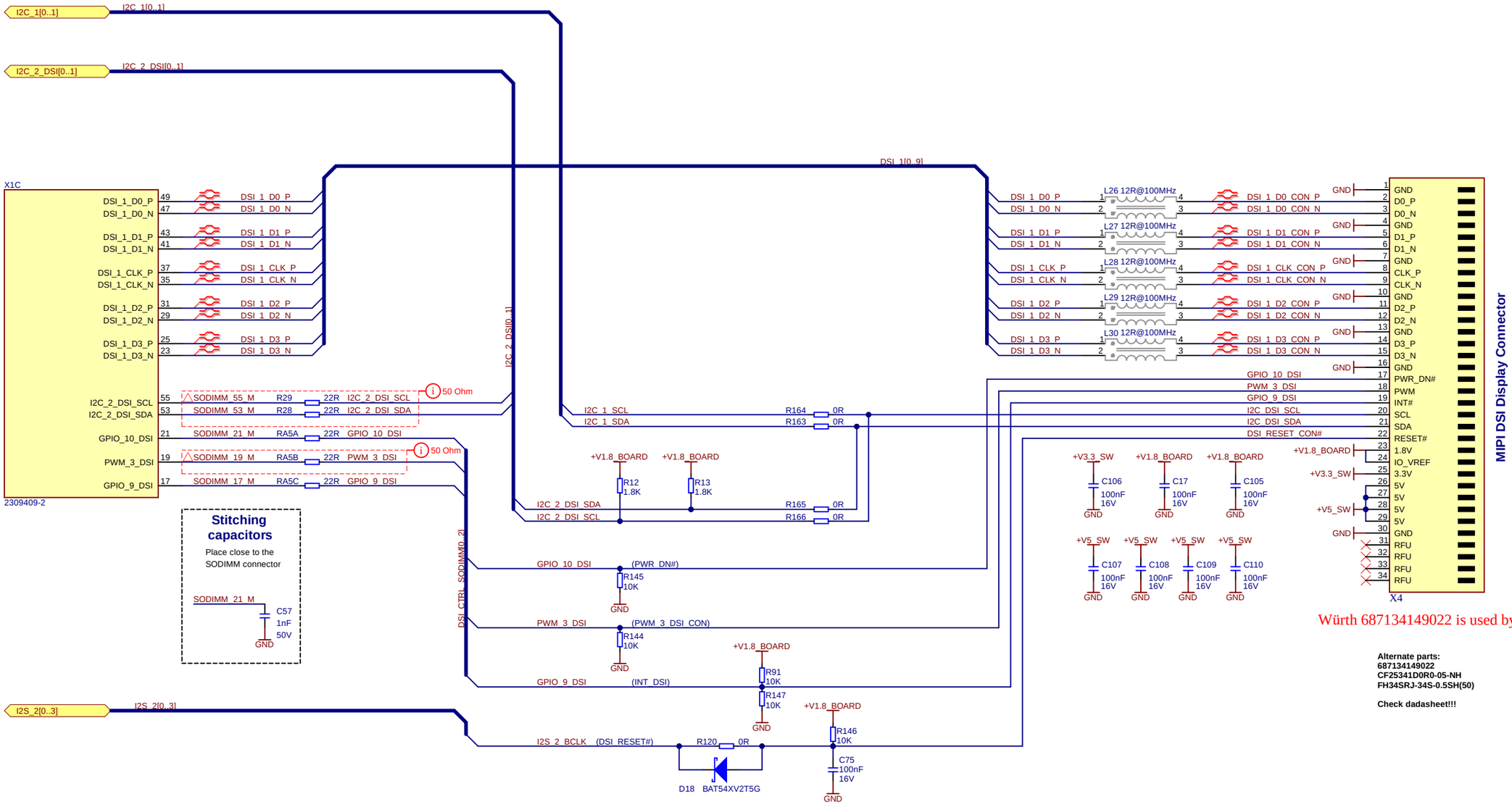
Title **cb-vis-ccu**		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number:9	Revision:V2.0
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 9 of 18	
File: Ethernet.SchDoc		



DIRx	Direction
0	A <= B
1	A => B

VIRGINIA INDUSTRY SOLUTIONS

Title *cb-vis-ccu*		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number: 10	Revision: V2.0
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 10 of 18	
File: MIPI_CSI.SchDoc		



Würth 687134149022 is used by deault

Alternate parts:
687134149022
CF25341D0R0-05-NH
FH34SRJ-34S-0.5SH(50)

Check dadasheet!!!

VIRGINIA INDUSTRY SOLUTIONS

Title	*cb-vis-ccu*		*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number: 11	Revision: V2.0	
Date: 10/15/2026 Time: 1:47:34 PM		Sheet 11 of 18	
File: MIPI_DSI.SchDoc			

I2C_3_HDMI[0..1]

+V3.3_SW

+V1.8_BOARD

D26
BAT54XV2T5G

R116
10K

R77
15K

R117
12K

T8B
NTZD3154NT1G

T8A
NTZD3154NT1G

D28
BAT54XV2T5G

D27
D5V0L1B2S9-7

GND

IC22

SN74LVC1G125DCKR

R118

R119 1.2K
47K

D29
D5V0L1B2S9-7

C116
100nF
16V

IC15

PCA9617ADPJ

C85
100nF
16V

R122
10K

R123
10K

R121
10K

C86
100nF
16V

R124
1.8K

R125
1.8K

HDMI_1_VDISP

I2C_3_HDMI_SCL

I2C_3_HDMI_SDA

HDMI_1_DDC_CON_SCL

HDMI_1_DDC_CON_SDA

D30
D5V0L1B2S9-7

D31
D5V0L1B2S9-7

50 Ohm

X1D

2309409-2

HDMI_1_CEC
HDMI_1_HPD
I2C_3_HDMI_SCL
I2C_3_HDMI_SDA
HDMI_1_TXD2_P
HDMI_1_TXD2_N
HDMI_1_TXD1_P
HDMI_1_TXD1_N
HDMI_1_TXD0_P
HDMI_1_TXD0_N
HDMI_1_TXC_P
HDMI_1_TXC_N

SODIMM_63_M_RA21A
SODIMM_61_M_RA21B
SODIMM_59_M_RA21C
SODIMM_57_M_RA21D

22R HDMI_1_CEC
22R HDMI_1_HPD
22R I2C_3_HDMI_SCL
22R I2C_3_HDMI_SDA

HDMI_1[0..7]

L17
L18
L19
L20

12R@100MHz

HDMI_1_TXD2_CON_P
HDMI_1_TXD2_CON_N
HDMI_1_TXD1_CON_P
HDMI_1_TXD1_CON_N
HDMI_1_TXD0_CON_P
HDMI_1_TXD0_CON_N
HDMI_1_TXC_CON_P
HDMI_1_TXC_CON_N

D32
DT1240-04LP-7

D33
DT1240-04LP-7

X14

HDMI 2.0 Connector

RAHHD19TR

TDMS_D2P
TDMS_D2_Shield
TDMS_D2N
TDMS_D1P
TDMS_D1_Shield
TDMS_D1N
TDMS_D0P
TDMS_D0_Shield
TDMS_D0N
TDMS_CLKP
TDMS_CLK_Shield
TDMS_CLKN
CEC
HEC
DDC_SCL
DDC_SDA
GND
PWR_5V_50mA
HotPlug_HECDP
SH1
SH2
SH3
SH4

HDMI_1_CEC_CON
HDMI_1_HPD_CON

D3
D5V0L1B2S9-7

R190
1M

C7
1nF
2KV

GND_CHASSIS

Alternate part number: HDMI-019S

+V5_SW

IC16

NCP380HMUAJAATBG

C88
100nF
16V

R126
10K

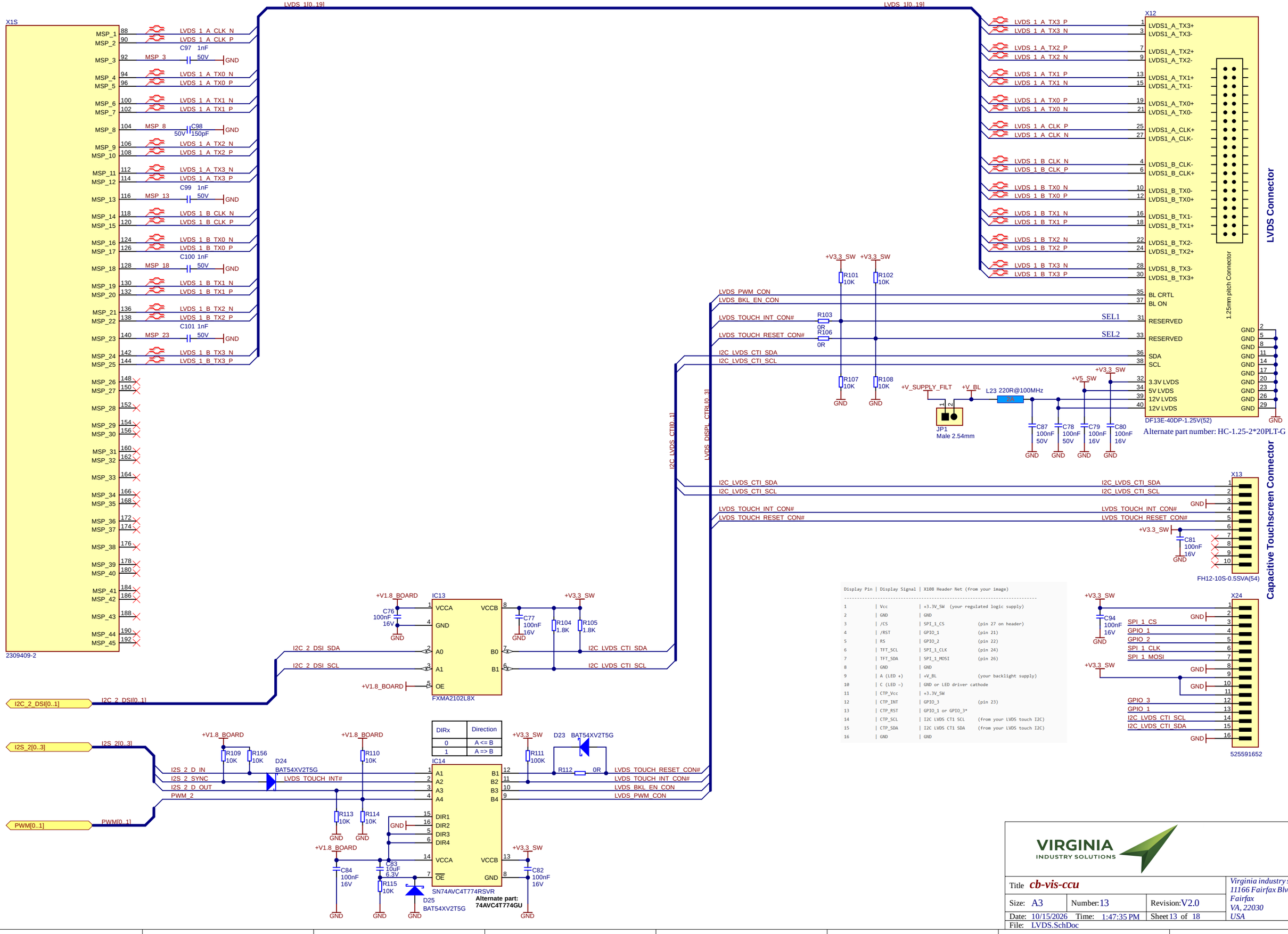
C89
1uF
16V

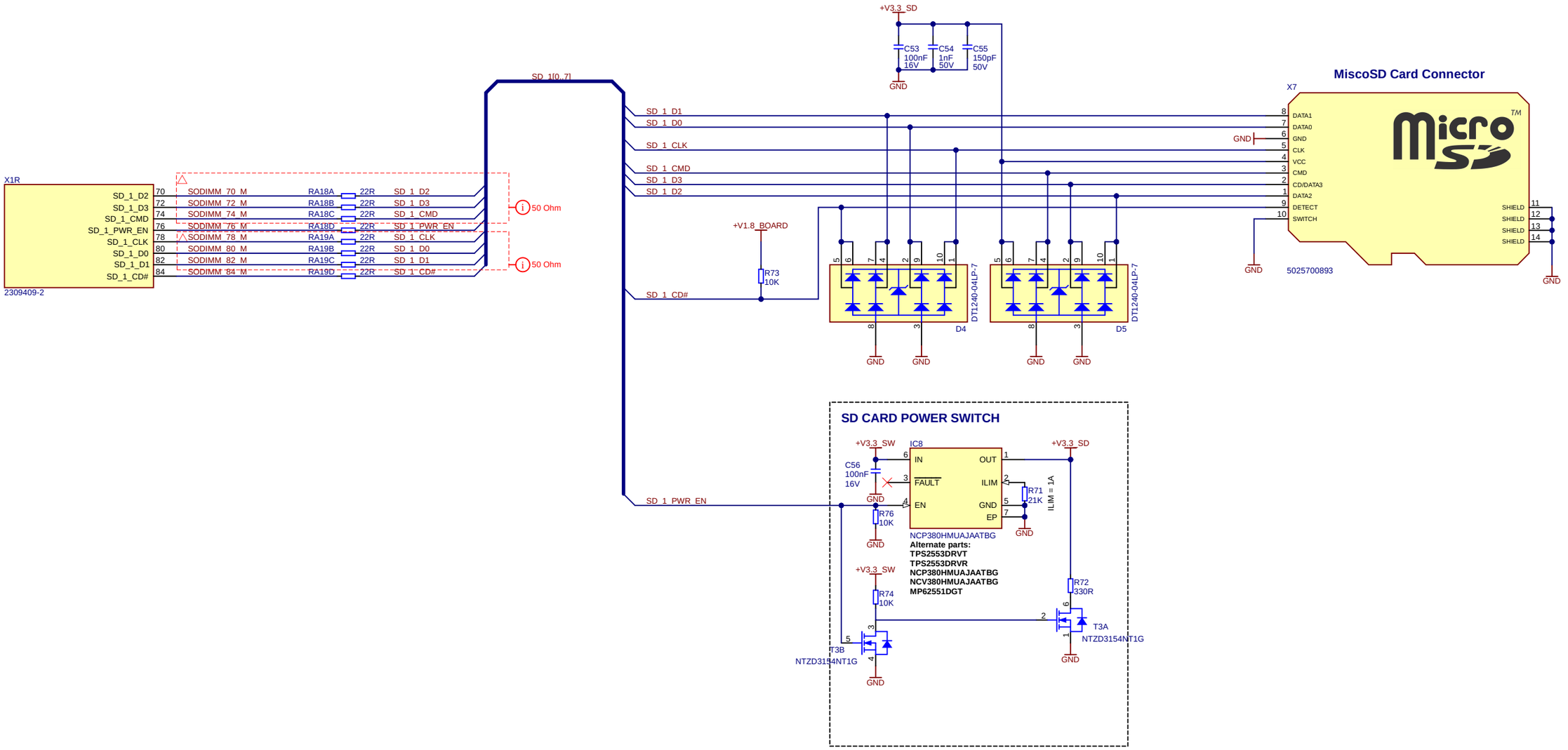
R127
100K

Alternate parts:
TPS2553DRVT
TPS2553DRVR
NCP380HMUAJAATBG
NCV380HMUAJAATBG
MP62551DGT

VIRGINIA INDUSTRY SOLUTIONS

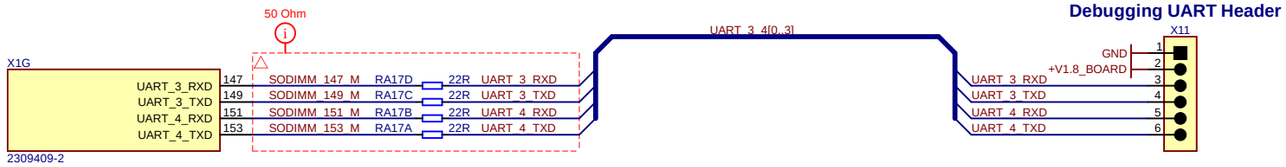
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Size: A3	Number: 12	Revision: V2.0
Date: 10/15/2026 Time: 1:47:34 PM	Sheet 12 of 18	
File: HDMI.SchDoc		


X1S
LVDS_1[0..19]
MSP_1 88 LVDS_1_A_CLK_N
MSP_2 90 LVDS_1_A_CLK_P
C97 1nF
MSP_3 92 MSP_3 50V GND
MSP_4 94 LVDS_1_A_TX0_N
MSP_5 96 LVDS_1_A_TX0_P
MSP_6 100 LVDS_1_A_TX1_N
MSP_7 102 LVDS_1_A_TX1_P
MSP_8 104 MSP_8 C98 50V 150pF GND
MSP_9 106 LVDS_1_A_TX2_N
MSP_10 108 LVDS_1_A_TX2_P
MSP_11 112 LVDS_1_A_TX3_N
MSP_12 114 LVDS_1_A_TX3_P
C99 1nF
MSP_13 116 MSP_13 50V GND
MSP_14 118 LVDS_1_B_CLK_N
MSP_15 120 LVDS_1_B_CLK_P
MSP_16 124 LVDS_1_B_TX0_N
MSP_17 126 LVDS_1_B_TX0_P
C100 1nF
MSP_18 128 MSP_18 50V GND
MSP_19 130 LVDS_1_B_TX1_N
MSP_20 132 LVDS_1_B_TX1_P
MSP_21 136 LVDS_1_B_TX2_N
MSP_22 138 LVDS_1_B_TX2_P
C101 1nF
MSP_23 140 MSP_23 50V GND
MSP_24 142 LVDS_1_B_TX3_N
MSP_25 144 LVDS_1_B_TX3_P
MSP_26 148
MSP_27 150
MSP_28 152
MSP_29 154
MSP_30 156
MSP_31 160
MSP_32 162
MSP_33 164
MSP_34 166
MSP_35 168
MSP_36 172
MSP_37 174
MSP_38 176
MSP_39 178
MSP_40 180
MSP_41 184
MSP_42 186
MSP_43 188
MSP_44 190
MSP_45 192
2309409-2
LVDS_1[0..19]
X12
LVDS_1_A_TX3_P 1 LVDS1_A_TX3+
LVDS_1_A_TX3_N 3 LVDS1_A_TX3-
LVDS_1_A_TX2_P 7 LVDS1_A_TX2+
LVDS_1_A_TX2_N 9 LVDS1_A_TX2-
LVDS_1_A_TX1_P 13 LVDS1_A_TX1+
LVDS_1_A_TX1_N 15 LVDS1_A_TX1-
LVDS_1_A_TX0_P 19 LVDS1_A_TX0+
LVDS_1_A_TX0_N 21 LVDS1_A_TX0-
LVDS_1_A_CLK_P 25 LVDS1_A_CLK+
LVDS_1_A_CLK_N 27 LVDS1_A_CLK-
LVDS_1_B_CLK_N 4 LVDS1_B_CLK-
LVDS_1_B_CLK_P 6 LVDS1_B_CLK+
LVDS_1_B_TX0_N 10 LVDS1_B_TX0-
LVDS_1_B_TX0_P 12 LVDS1_B_TX0+
LVDS_1_B_TX1_N 16 LVDS1_B_TX1-
LVDS_1_B_TX1_P 18 LVDS1_B_TX1+
LVDS_1_B_TX2_N 22 LVDS1_B_TX2-
LVDS_1_B_TX2_P 24 LVDS1_B_TX2+
LVDS_1_B_TX3_N 28 LVDS1_B_TX3-
LVDS_1_B_TX3_P 30 LVDS1_B_TX3+
BL CRTL 35
BL ON 37
SEL1 31 RESERVED
SEL2 33 RESERVED
SDA 36
SCL 38
3.3V LVDS 32
5V LVDS 34
12V LVDS 39
12V LVDS 40
GND 2 5 8 11 14 17 20 23 26 29
1.25mm pitch Connector
LVDS Connector
DF13E-40DP-1.25V(52)
Alternate part number: HC-1.25-2*20PLT-G
+V3.3_SW
R101 10K
R102 10K
LVDS_PWM_CON
LVDS_BKL_EN_CON
LVDS_TOUCH_INT_CON# R103 0R
LVDS_TOUCH_RESET_CON# R106 0R
I2C_LVDS_CTI_SDA
I2C_LVDS_CTI_SCL
R107 10K
R108 10K
+V_SUPPLY_FILT
+V_BL
L23 220R@100MHz
+V5_SW
JP1
Male 2.54mm
C87 100nF 50V
C78 100nF 50V
C79 100nF 16V
C80 100nF 16V
I2C_LVDS_CTI[0..1]
LVDS_DISPL_CTRL[0..3]
I2C_LVDS_CTI_SDA
I2C_LVDS_CTI_SCL
LVDS_TOUCH_INT_CON#
LVDS_TOUCH_RESET_CON#
X13
C81 100nF 16V
FH12-10S-0.5SVA(54)
Capacitive Touchscreen Connector
Display Pin | Display Signal | X108 Header Net (from your image)
1 | Vcc | +3.3V_SW (your regulated logic supply)
2 | GND | GND
3 | /CS | SPI_1_CS (pin 27 on header)
4 | /RST | GPIO_1 (pin 21)
5 | RS | GPIO_2 (pin 22)
6 | TFT_SCL | SPI_1_CLK (pin 24)
7 | TFT_SDA | SPI_1_MOSI (pin 26)
8 | GND | GND
9 | A (LED +) | +V_BL (your backlight supply)
10 | C (LED −) | GND or LED driver cathode
11 | CTP_Vcc | +3.3V_SW
12 | CTP_INT | GPIO_3 (pin 23)
13 | CTP_RST | GPIO_1 or GPIO_3*
14 | CTP_SCL | I2C LVDS CT1 SCL (from your LVDS touch I2C)
15 | CTP_SDA | I2C LVDS CT1 SDA (from your LVDS touch I2C)
16 | GND | GND
X24
C94 100nF 16V
SPI_1_CS
GPIO_1
GPIO_2
SPI_1_CLK
SPI_1_MOSI
GPIO_3
GPIO_1
I2C_LVDS_CTI_SCL
I2C_LVDS_CTI_SDA
525591652
+V1.8_BOARD
IC13
C76 100nF 16V
C77 100nF 16V
R104 1.8K
R105 1.8K
I2C_2_DSI_SDA
I2C_2_DSI_SCL
I2C_LVDS_CTI_SDA
I2C_LVDS_CTI_SCL
FXMA2102L8X
I2C_2_DSI[0..1]
I2S_2[0..3]
PWM[0..1]
R109 10K
R156 10K
R110 10K
D24 BAT54XV2T5G
I2S_2_D_IN
I2S_2_SYNC
I2S_2_D_OUT
PWM_2
LVDS_TOUCH_INT#
R111 100K
D23 BAT54XV2T5G
R112 0R
LVDS_TOUCH_RESET_CON#
LVDS_TOUCH_INT_CON#
LVDS_BKL_EN_CON
LVDS_PWM_CON
R113 10K
R114 10K
DIRx | Direction
0 | A <= B
1 | A => B
IC14
C84 100nF 16V
C83 10uF 6.3V
R115 10K
C82 100nF 16V
D25 BAT54XV2T5G
SN74AVC4T774RSVR
Alternate part: 74AVC4T774GU
VIRGINIA INDUSTRY SOLUTIONS
Title cb-vis-ccu
Size: A3
Number: 13
Revision: V2.0
Date: 10/15/2026 Time: 1:47:35 PM
Sheet 13 of 18
File: LVDS.SchDoc
Virginia industry solutions
11166 Fairfax Blvd
Fairfax
VA, 22030
USA



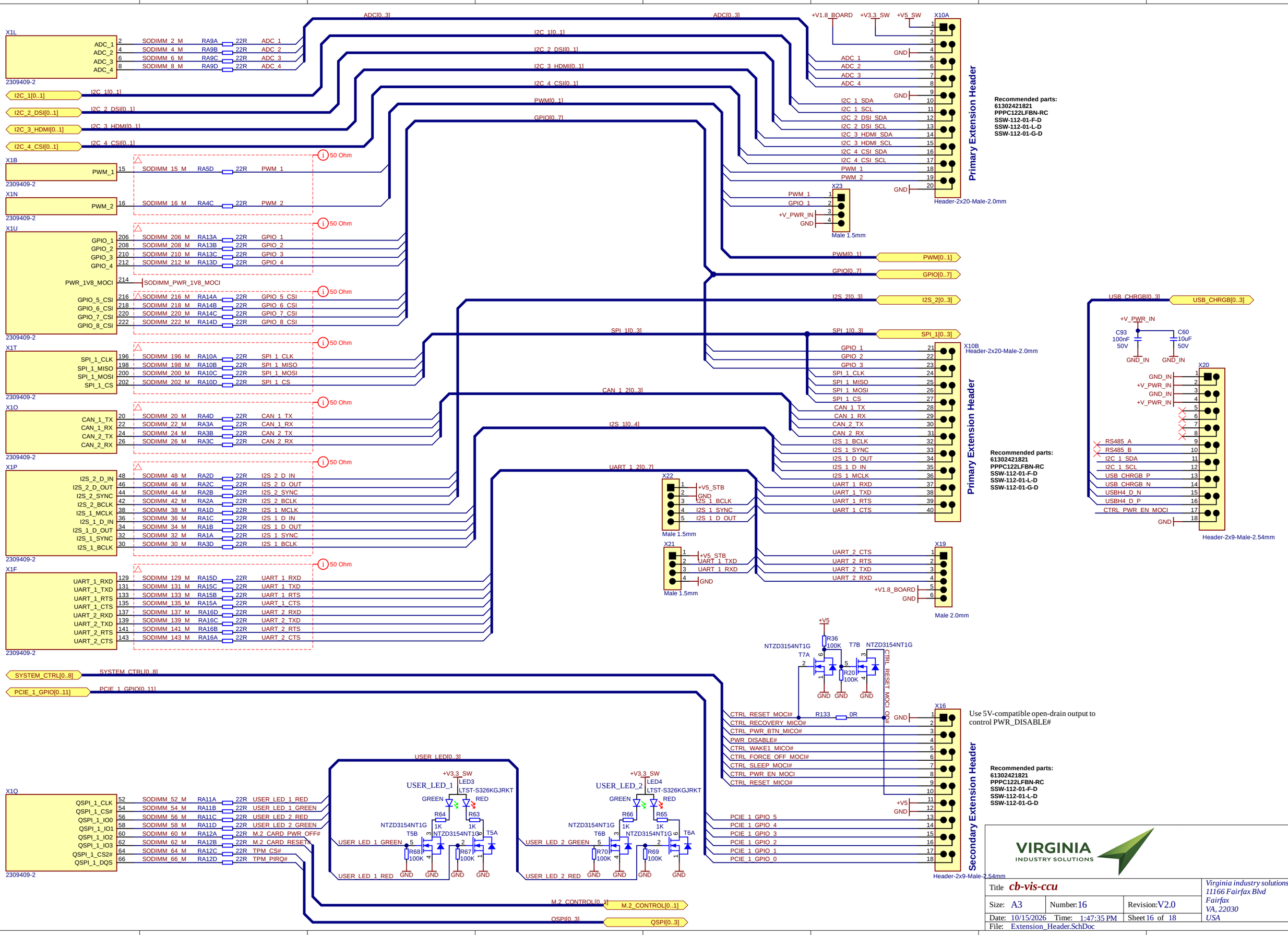
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Size: A3	Number: 14	Revision: V2.0	
Date: 10/15/2026 Time: 1:47:35 PM		Sheet 14 of 18	
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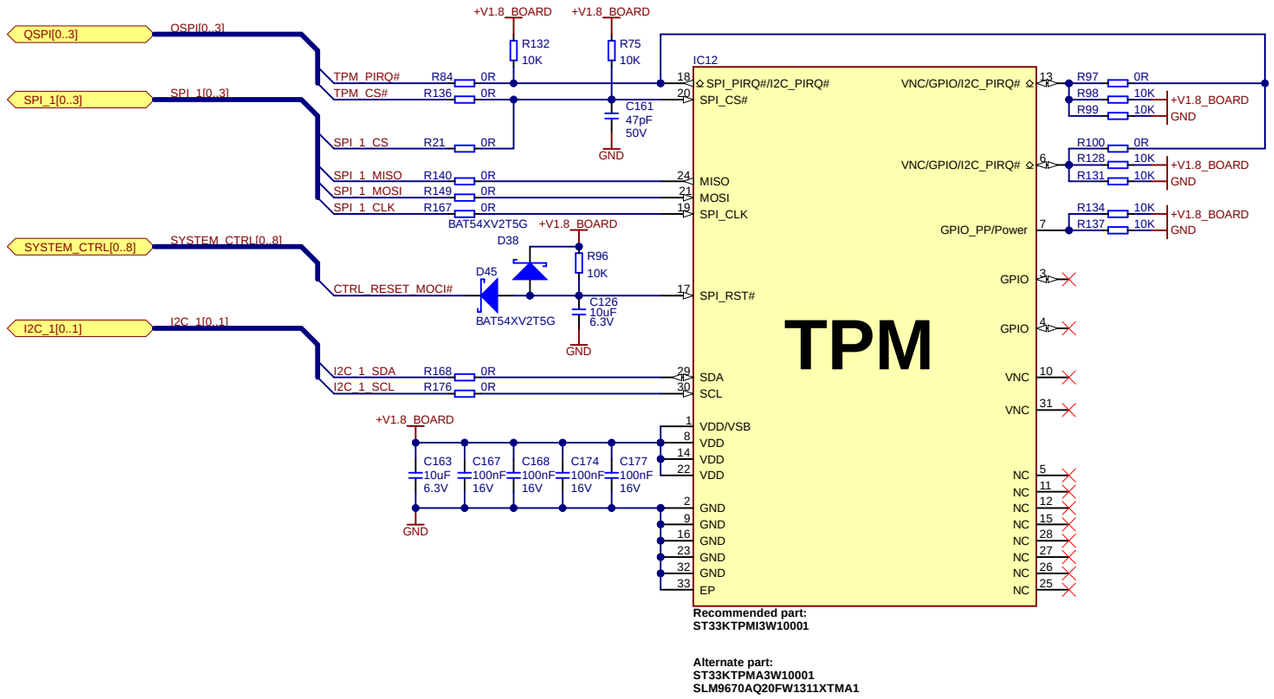




VIRGINIA INDUSTRY SOLUTIONS

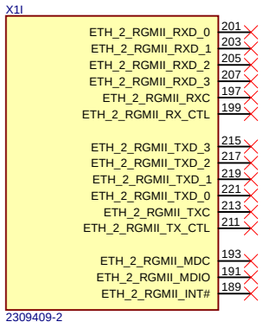
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Size: A3	Number:15	Revision:V2.0	
Date: 10/15/2026 Time: 1:47:35 PM		Sheet 15 of 18	
File: Debug.SchDoc			


ADC[0..3]
I2C_1[0..1]
I2C_2_DSI[0..1]
I2C_3_HDMI[0..1]
I2C_4_CSI[0..1]
PWM[0..1]
GPIO[0..7]
SPI_1[0..3]
CAN_1_2[0..3]
I2S_1[0..4]
UART_1_2[0..7]
SYSTEM_CTRL[0..8]
PCIE_1_GPIO[0..11]
USER_LED[0..3]
M.2_CONTROL[0..1]
QSPI[0..3]
USB_CHRGB[0..3]
X1L
ADC_1
ADC_2
ADC_3
ADC_4
2309409-2
SODIMM_2_M RA9A 22R ADC_1
SODIMM_4_M RA9B 22R ADC_2
SODIMM_6_M RA9C 22R ADC_3
SODIMM_8_M RA9D 22R ADC_4
50 Ohm
X1B
PWM_1
SODIMM_15_M RA5D 22R PWM_1
X1N
PWM_2
SODIMM_16_M RA4C 22R PWM_2
X1U
GPIO_1
GPIO_2
GPIO_3
GPIO_4
PWR_1V8_MOCI
GPIO_5_CSI
GPIO_6_CSI
GPIO_7_CSI
GPIO_8_CSI
SODIMM_206_M RA13A 22R GPIO_1
SODIMM_208_M RA13B 22R GPIO_2
SODIMM_210_M RA13C 22R GPIO_3
SODIMM_212_M RA13D 22R GPIO_4
SODIMM_PWR_1V8_MOCI
SODIMM_216_M RA14A 22R GPIO_5_CSI
SODIMM_218_M RA14B 22R GPIO_6_CSI
SODIMM_220_M RA14C 22R GPIO_7_CSI
SODIMM_222_M RA14D 22R GPIO_8_CSI
X1T
SPI_1_CLK
SPI_1_MISO
SPI_1_MOSI
SPI_1_CS
SODIMM_196_M RA10A 22R SPI_1_CLK
SODIMM_198_M RA10B 22R SPI_1_MISO
SODIMM_200_M RA10C 22R SPI_1_MOSI
SODIMM_202_M RA10D 22R SPI_1_CS
X1O
CAN_1_TX
CAN_1_RX
CAN_2_TX
CAN_2_RX
SODIMM_20_M RA4D 22R CAN_1_TX
SODIMM_22_M RA3A 22R CAN_1_RX
SODIMM_24_M RA3B 22R CAN_2_TX
SODIMM_26_M RA3C 22R CAN_2_RX
X1P
I2S_2_D_IN
I2S_2_D_OUT
I2S_2_SYNC
I2S_2_BCLK
I2S_1_MCLK
I2S_1_D_IN
I2S_1_D_OUT
I2S_1_SYNC
I2S_1_BCLK
SODIMM_48_M RA2D 22R I2S_2_D_IN
SODIMM_46_M RA2C 22R I2S_2_D_OUT
SODIMM_44_M RA2B 22R I2S_2_SYNC
SODIMM_42_M RA2A 22R I2S_2_BCLK
SODIMM_38_M RA1D 22R I2S_1_MCLK
SODIMM_36_M RA1C 22R I2S_1_D_IN
SODIMM_34_M RA1B 22R I2S_1_D_OUT
SODIMM_32_M RA1A 22R I2S_1_SYNC
SODIMM_30_M RA3D 22R I2S_1_BCLK
X1F
UART_1_RXD
UART_1_TXD
UART_1_RTS
UART_1_CTS
UART_2_RXD
UART_2_TXD
UART_2_RTS
UART_2_CTS
SODIMM_129_M RA15D 22R UART_1_RXD
SODIMM_131_M RA15C 22R UART_1_TXD
SODIMM_133_M RA15B 22R UART_1_RTS
SODIMM_135_M RA15A 22R UART_1_CTS
SODIMM_137_M RA16D 22R UART_2_RXD
SODIMM_139_M RA16C 22R UART_2_TXD
SODIMM_141_M RA16B 22R UART_2_RTS
SODIMM_143_M RA16A 22R UART_2_CTS
X1Q
QSPI_1_CLK
QSPI_1_CS#
QSPI_1_IO0
QSPI_1_IO1
QSPI_1_IO2
QSPI_1_IO3
QSPI_1_CS2#
QSPI_1_DQS
SODIMM_52_M RA11A 22R USER_LED_1_RED
SODIMM_54_M RA11B 22R USER_LED_1_GREEN
SODIMM_56_M RA11C 22R USER_LED_2_RED
SODIMM_58_M RA11D 22R USER_LED_2_GREEN
SODIMM_60_M RA12A 22R M.2_CARD_PWR_OFF#
SODIMM_62_M RA12B 22R M.2_CARD_RESET#
SODIMM_64_M RA12C 22R TPM_CS#
SODIMM_66_M RA12D 22R TPM_PIRQ#
+V1.8_BOARD
+V3.3_SW
+V5_SW
X10A
Primary Extension Header
Header-2x20-Male-2.0mm
Recommended parts:
61302421821
PPPC122LFBN-RC
SSW-112-01-F-D
SSW-112-01-L-D
SSW-112-01-G-D
X23
Male 1.5mm
+V_PWR_IN
X10B
Header-2x20-Male-2.0mm
X22
+V5_STB
Male 1.5mm
X21
Male 1.5mm
X19
Male 2.0mm
+V5
NTZD3154NT1G
T7A
T7B
R36
100K
R20
100K
CTRL_RESET_MOCI_OD#
R133
0R
X16
Secondary Extension Header
Header-2x9-Male-2.54mm
Use 5V-compatible open-drain output to control PWR_DISABLE#
USER_LED_1
USER_LED_2
LED3
LED4
LTST-S326KGJRKT
GREEN
RED
R64
R63
R66
R65
1K
T5B
T5A
T6B
T6A
R68
R67
R70
R69
100K
C93
100nF
50V
C60
10uF
50V
GND_IN
X20
Header-2x9-Male-2.54mm
RS485_A
RS485_B
I2C_1_SDA
I2C_1_SCL
USB_CHRGB_P
USB_CHRGB_N
USBH4_D_N
USBH4_D_P
CTRL_PWR_EN_MOCI
VIRGINIA
INDUSTRY SOLUTIONS
Title cb-vis-ccu
Size: A3
Number: 16
Revision: V2.0
Date: 10/15/2026
Time: 1:47:35 PM
Sheet 16 of 18
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Virginia industry solutions
11166 Fairfax Blvd
Fairfax
VA, 22030
USA

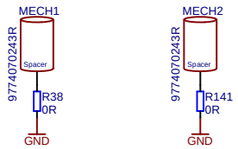


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Date: 10/15/2026 Time: 1:47:35 PM		Sheet 17 of 18	
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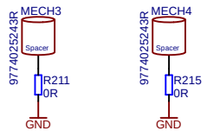
UNUSED PARTS



HEATSINK STANDOFFS



SOM STANDOFFS



M.2 CARD STANDOFFS



Alternate part number: SMTWE2010MTJ

LVDS POWER JUMPER



PCB MOUNTING HOLES









VIRGINIA INDUSTRY SOLUTIONS

Title **cb-vis-ccu**			*Virginia industry solutions* *11166 Fairfax Blvd* *Fairfax* *VA, 22030* *USA*
Size: A3	Number: 17	Revision: V2.0	
Date: 10/15/2026 Time: 1:47:35 PM		Sheet 18 of 18	
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